

KEYONAI WILLIAMS

Computer Vision, Autonomy & Navigation | AI/ML Intern Candidate
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PROFESSIONAL SUMMARY

Computer Science student (AI & Machine Learning focus) with hands-on experience building real-time computer vision and perception systems in C++ and Python, including object detection, monocular depth estimation, and optical flow for motion tracking. Currently learning ROS2 for autonomous robotics applications. Strong interest in navigation, autonomy, and perception for unmanned systems operating in GNSS-contested environments, with hands-on industrial robotics exposure from operating automated assembly systems in a high-precision manufacturing environment.

TECHNICAL SKILLS

- Languages: Python | C++17 | SQL (MySQL)
- Robotics: ROS2 (In Progress)
- Computer Vision & Perception: YOLOv8 | MiDaS Depth Estimation | Optical Flow (RAFT) | OpenCV | ONNX | SAM (Segment Anything) | OCR (Tesseract, PaddleOCR, EasyOCR) | CMake
- ML & Data: PyTorch | Scikit-learn | LLMs | RAG Pipelines | LlamaIndex | NLP | NumPy | Pandas | Matplotlib | Git | Linux

EXPERIENCE

Pfizer — AI-Powered Document Insights & Data Extraction

May 2026 – Present

Extern, Remote

- Built Python-based data processing pipelines to extract and structure data from complex documents, delivering production-ready automation with OCR and computer vision techniques.
- Benchmarked TesseractOCR, PaddleOCR, and EasyOCR across confidence, detection rate, and accuracy to support engineering decisions — demonstrating rigorous algorithm testing and assessment methodology.
- Developed a RAG pipeline using LlamaIndex and vector databases; deployed a Gradio-based chatbot interface for AI-powered document Q&A.

AI Trainer — Handshake AI

Feb 2026 – May 2026

Remote

- Trained and evaluated multimodal AI models across audio, visual, and text data; validated outputs, documented findings, and maintained quality benchmarks.

Body Weld Team Member — Mazda Toyota Manufacturing

Aug 2023 – Present

Madison, AL

- Operate automated welding and robotic assembly systems in a high-precision manufacturing environment, gaining hands-on exposure to industrial robotics, sensor/hardware integration, and mission-critical production systems.

RELEVANT PROJECTS

YOLO + MiDaS Depth-Aware Object Detection | C++17, YOLOv8, MiDaS, OpenCV, ONNX, CMake

- Built a real-time C++ perception system combining YOLOv8 object detection with MiDaS monocular depth estimation to estimate object distance from a single RGB camera, without a dedicated depth sensor — directly applicable to vision-based navigation.

Optical Flow Visualization | C++, RAFT, LibTorch, OpenCV

- Implemented real-time optical flow using RAFT-small via LibTorch for motion direction and speed visualization — a core technique for vision-aided navigation and state estimation in GNSS-degraded environments.

SAM Image Segmentation | Python, SAM, Ultralytics, OpenCV, NumPy

- Implemented Meta's Segment Anything Model for point-prompted object segmentation across images, webcam feeds, and video — demonstrating scene understanding applicable to simulation and autonomy testbeds.

RAG Document Intelligence Pipeline | Python, LlamaIndex, Vector DBs, Gradio, OCR

- Built a complete RAG pipeline enabling AI-powered semantic search and Q&A over large document sets, with a Gradio chatbot interface.

EDUCATION

B.S. Computer Science — AI & Machine Learning Focus

Expected May 2027

Western Governors University (ABET-Accredited)

- Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, C++, SQL, Git, Network & Security Foundations, Web Development, Probability & Statistics